

The one-variable density for unequal exponents

Claude, with Daniel Murfet

2026-09-07

Abstract

Unit 86 gave the exact one-variable density of the $d = 2$ chart integral $\iint u^{h_1} v^{h_2} e^{-\beta(Nu^{k_1}v^{k_2})^2} \Psi(u, Nu^{k_1}v^{k_2}) du dv$ when the two exponents $(h_i + 1)/k_i$ agree. Here the restriction is dropped: with $p_2 = (h_2 + 1)/k_2$ the density in $r = u^{k_1}v^{k_2}$ is

$$\rho(r, s) = \frac{1}{k_2} r^{p_2-1} \int_{(r/b^{k_2})^{1/k_1}}^b u^{h_1-k_1p_2} \Psi(u, s) du,$$

a weighted axis integral with weight $u^{-1-\gamma}$, $\gamma = k_1(p_2 - p_1)$. This is exactly the shape consumed by the weighted finite-part axis lemma (unit 100): the face terms $u^i a_i(v, s)$ of the rectangular decomposition have exponents $p + i/k_1 \neq p$, and their densities are now available. First step of Astra #4's post-rank-5 programme (monomial block densities and the τ -cutoff theorem). Zero sorry/axiom.

1 The things formalised

```

noncomputable def genDensity (p b : ℝ) (h k k : ℝ) (Ψ : ℝ → ℝ) (r s : ℝ) : ℝ :=
  1 / (k : ℝ) * r ^ (p - 1)
  * u in Icc ((r / b ^ k) ^ ((k : ℝ) ^ 1)) b, u ^ ((h : ℝ) - k * p) * Ψ u s
noncomputable def regionIntegrand' (b N p : ℝ) (h k k : ℝ) (Ψ : ℝ → ℝ) (u r : ℝ) :
theorem regionIntegrand'_measurable / _integral_r / _integral_u / _integrable
theorem rpow_weight_mul_pow (u p : ℝ) (h k : ℝ) (hu : 0 < u) :
  u ^ ((h : ℝ) - k * p) * (u ^ k) ^ p = u ^ h
theorem twoDamp_uOnly_eq_transferZ' (b N p : ℝ) (h h k k : ℝ) (h : 0 < ) (hb : 0 < b)
(hN : 0 < N) (hk : 0 < k) (hk : 0 < k) (hp : ((h : ℝ) + 1) / k = p)
(Ψ : ℝ → ℝ) (hΨc : Continuous (Function.uncurry Ψ)) :
twoDamp b N h h k k (fun u _ s => Ψ u s)
= transferZ (genDensity p b h k k Ψ) (b ^ (k + k)) N
theorem genDensity_eq_axis (p b : ℝ) (h k k : ℝ) (Ψ : ℝ → ℝ) (r s : ℝ) :
genDensity p b h k k Ψ r s
= 1 / (k : ℝ) * r ^ (p - 1)
  * u in Ioc ((r / b ^ k) ^ ((k : ℝ) ^ 1)) b,
  u ^ (-1 - ((k : ℝ) * p - h - 1)) * Ψ u s

```

2 Mathematics

The proof is the unit-86 proof with one change of weight. The inner substitution $r = u^{k_1}v^{k_2}$ turns $\int_0^b v^{h_2} g(u^{k_1}v^{k_2}) dv$ into $k_2^{-1}(u^{k_1})^{-p_2} \int_0^{u^{k_1}b^{k_2}} r^{p_2-1} g(r) dr$, and the outer weight combines as $u^{h_1}(u^{k_1})^{-p_2} = u^{h_1-k_1p_2}$ (in unit 86 this was u^{-1}). Fubini over the curved region $\{r \leq u^{k_1}b^{k_2}\}$

then gives the u -section $[(r/b^{k_2})^{1/k_1}, b]$. Integrability of the region integrand uses the bound $u^{h_1-k_1p_2}(u^{k_1}b^{k_2})^{p_2} = b^{k_2p_2}u^{h_1}$, integrable on $(0, b]$ since $h_1 \geq 0$ (`rpow_weight_mul_pow`).

Writing $-1 - \gamma = h_1 - k_1p_2$, i.e. $\gamma = k_1p_2 - h_1 - 1 = k_1(p_2 - p_1)$, the inner integral is $\int_\varepsilon^b u^{-1-\gamma} \Psi(u, s) du$ with $\varepsilon = (r/b^{k_2})^{1/k_1}$ (`genDensity_eq_axis`), so the finite-part lemma applies verbatim with $A = b^{k_2}$, $k = k_1$.

3 Paper-facing meaning

In the rectangular decomposition of the $d = 2$ block, a face term $u^i a_i(v, s)$ is (after `twoDAmp_swap`) the case $\Psi = a_i$ with weights v^{h_2}, u^{h_1+i} ; its density has $p_2 \rightarrow p + i/k_1$ and $\gamma = k_2i/k_1$, so unit 100 produces the exponents $p + m/k_2$, a logarithm exactly at the collision $i/k_1 = m/k_2$, and the finite-part coefficient. The next units assemble these into the cutoff-indexed density expansion and feed it to the transfer theorem.

4 Lean notes

- `open Finset` together with `open Set` makes `mem_Ioc`, `mem_Iic`, `mem_Ici`, `mem_Icc` ambiguous in `simp` only; open only `Set` in measure-theoretic files.
- `integral_Icc_eq_integral_Ioc` inside `rw` leaves `ClosedIicTopology ?m` stuck even with `(x := ...)` `(y := ...)`; and the same stuck instance reappears for `measurableSet_Ioc` after `congr`. Bind the lemma with all of `( := volume)` `(f := ...)` `(x := ...)` `(y := ...)` in a `have` and finish with a `show ... by ring` rewrite of the exponent rather than `congr`.
- `Real.rpow_natCast` in the forward direction converts $u^{\wedge (h : \mathbb{N})}$ to $u^{\wedge h}$ once the exponent identity is established.